AMITY UNIVERSITY
PATNA

Course Curriculum

Course Title: **Analytical Data Visualization**

Course Code: **CSIT394**

Credit Units: 5

Course Level: UG

Syllabus Version-I

L	T	P/S	SW	No. of PSDA	Total Credit Unit
3	0	4	0	0	5

Course Description:

This course introduces students to the concepts, principles, tools, and applications of data visualization and analytics. Students learn how to collect, prepare, analyze, interpret, and visualize data for effective decision-making. The course covers exploratory data analysis, visualization principles, dashboard development using Power BI, and ethical aspects of data analytics through hands-on activities and real-world case studies.

Course Objectives

CO1: Understand the fundamentals of data analytics and data visualization.

CO2: Analyse different types of data using exploratory data analysis and statistical techniques.

CO3: Apply visualization principles and appropriate chart selection for effective communication.

CO4: Evaluate ethical issues, data quality, and visualization effectiveness.

CO5: Design interactive dashboards and analytical reports using Power BI.

Pre-requisite

Students should have,Basic computer knowledge ,MS Excel familiarity ,Basic statistics (Mean, Median, Mode) ,Basic programming concepts ,Understanding of datasets and spreadsheets.

Course Contents/Syllabus:

Module-I	Weightage (%)	Lecture Hours
Introduction to Data Analytics and Data Visualization Introduction to Data ,Types of Data • Numerical • Categorical • Temporal	25	11

• Geographical Introduction to Data Analytics Types of Analytics • Descriptive Analytics • Diagnostic Analytics • Predictive Analytics • Prescriptive Analytics Introduction to Data Visualization ,Importance of Data Visualization ,History of Data Visualization ,Data-driven Decision Making ,Data Analytics Process ,Data Collection,Data Cleaning ,Data Preparation ,Analysis ,Visualization ,Interpretation ,Introduction to Visual Perception ,Reports and Dashboards ,Challenges and Limitations .		
Module-II	25	11
Exploratory Data Analysis and Visualization Tools,Introduction to Exploratory Data Analysis (EDA) ,Data Quality Assessment ,Missing Values ,Duplicate Records ,Outlier Detection ,Data Cleaning Techniques ,Data Transformation ,Summary Statistics ,Mean ,Median ,Mode ,Variance ,Standard Deviation ,Data Relationships ,Ranking ,Comparison ,Correlation ,Distribution ,Trends ,Introduction to Visualization Tools ,MS Excel ,Tableau ,Power BI ,Python (Matplotlib and Seaborn overview) ,Importing and Preparing Data in Power BI		
Module-III	25	12
Principles of Visualization Design Principles of Effective Visualization ,Choosing Appropriate Charts ,Visual Encoding ,Color Theory ,Dashboard Design Principles ,Interactive Dashboards ,KPI Design ,Data Storytelling ,Dashboard Layout ,Filters ,Slicers ,Drill-down ,Drill-through ,Best Practices,Ethical Data Visualization ,Misleading Graphs ,Data Privacy ,Responsible Data Usage ,Professional Ethics ,Evaluation of Visualization Quality .		
Module-IV	25	11
Analytical Thinking ,Business Intelligence Concept, KPI Analysis ,Sales Analytics , Customer Analytics , Education Analytics , Healthcare Analytics ,Financial Analytics , ,Interactive Dashboard Development using Power BI , Dashboard Publishing, Interpretation of Insights,Presentation of Analytical Reports ,Industry Applications , Mini Project		

Course Learning Outcomes (CLOs):

CLO1: Understand:
Explain the fundamental concepts, principles, and importance of data visualization and its role in data analysis and decision making.

CLO2: Analyze:
Analyze visual perception principles and evaluate their impact on effective data interpretation and communication.

CLO3: Evaluate:
Examine the ethical considerations, limitations, and challenges associated with data visualization and responsible use of data.

CLO4: Apply:
Apply appropriate visualization techniques and communicate insights effectively using different types of visual representations.

CLO5: Create:
Create, customize, and manage interactive dashboards and visualization projects using Power BI for practical applications.

Pedagogy for Course Delivery:

The course will be delivered through a combination of interactive lectures, demonstrations of visualization tools (such as Excel, Tableau, Power BI, and Python), real-world examples, case studies, guided hands-on activities, and practical exercises to develop understanding of data visualization concepts and their applications. Dashboard development, data interpretation, and mini project activities will be incorporated to encourage analytical thinking, communication, and practical problem-solving skills.

Assessment/ Examination Scheme:

Theory L/T (%)	Lab/Practical/Studio (%)	TOTAL
60	40	100

Theory Assessment (L&T):

Continuous Assessment/Internal Assessment (40 %)					End Term Examination (60%)
Components (Drop down)	Viva	Class Test	Home Assignment	Attendance	End Term
Weightage (%)	5	20	10	5	60

Lab/ Practical/ Studio Assessment:

Continuous Assessment/Internal Assessment (40 %)						End Term Examination (60%)	
Components (Drop down)	Class Test (Practical based)	Lab Record	Continuous Performance	Internal Viva	Attendance	Practical Evaluation	Viva
Weightage (%)	10	10	10	5	5	40	20

Data Visualization: Laboratory Experiments [60 Hr.]

S.No.	Experiment	Practical Hours
1	Create basic visualizations including Bar Chart, Column Chart, Pie Chart, Histogram, and Line Chart using Excel/Python/Power BI. Compare their applications.	4
2	Import datasets into Excel, Python (Pandas), and Power BI. Perform data inspection and understand dataset structure, variables, and data types.	4
3	Perform data preparation by handling missing values, removing duplicates, correcting inconsistent data, formatting data types, and transforming variables.	4
4	Perform Exploratory Data Analysis (EDA) using descriptive statistics (mean, median, mode, variance, standard deviation, quartiles, range, and frequency distribution).	4
5	Identify outliers using Box Plot and IQR method. Analyze data distribution using Histograms and Density Plots.	4
6	Create advanced visualizations including Scatter Plot, Bubble Chart, Box Plot, Heat Map, Tree map, and Area Chart. Interpret insights obtained from each visualization.	4
7	Perform correlation analysis using correlation matrix and Heat Map. Identify relationships among variables and interpret positive and negative correlations.	4
8	Perform ranking, comparison, deviation, and contribution analysis using suitable charts (Top-N, Waterfall, Funnel, Pareto, etc.).	4
9	Perform time-series analysis using Line Charts, Running Totals, Moving Average, Trend Lines, and Forecasting techniques	4
10	Design interactive dashboards in Power BI using filters, slicers, bookmarks, drill-down, drill-through, and KPI cards.	4
11	Build a business analytics dashboard using a real-world dataset (Sales, Healthcare, Education, Finance, E-commerce, etc.) and generate actionable insights.	4

12	Develop a complete data storytelling presentation based on dashboard findings, highlighting trends, patterns, and business recommendations.	4
13	Evaluate misleading visualizations, identify visualization errors and biases, redesign dashboards using ethical visualization principles, and justify improvements.	4
14	Mini Project: Design and implement an end-to-end analytics solution involving data preparation, EDA, visualization, dashboard creation, and presentation using a real-world dataset.	4
15	Prepare a data storytelling presentation based on dashboard findings, highlighting trends, patterns, and recommendations for decision-making.	4

Mapping Continuous Evaluation components/PSDA with CLOs					
SN	Bloom's Level	Course Learning Outcomes	Class Test	Home Assignment	Viva
1	Understand	Explain the fundamental concepts, principles, and importance of data visualization and its role in data analysis and decision making.	✓	✓	✓
2	Analyze	Analyze visual perception principles and evaluate their impact on effective data interpretation and communication.	✓	✓	✓
3	Evaluate	Examine the ethical considerations, limitations, and challenges associated with data visualization and responsible use of data.	✓	✓	✓
4	Apply	Apply appropriate visualization techniques and communicate insights effectively using different types of visual representations.		✓	✓
5	Create	Create, customize, and manage interactive dashboards and visualization projects using Power BI for practical applications.		✓	✓

Textbooks:

Chun-houh Chen, Wolfgang Härdle, Antony Unwin, "Handbook of Data Visualization", Springer, 2008.

Pérez, J. and Vialcanet, G., Visualize It: A Comprehensive Guide to Data Visualization, 2013.

Storytelling with Data: A Data Visualization Guide for Business Professionals Cole Nussbaumer Knaflic, Wiley; 1st edition, 2015.

The Visual Display of Quantitative Information by Edward Tufte, Graphics Press USA; 2nd edition, 2001.

Reference books:

1. Data Visualization: A Practical Introduction Kieran Healy, Princeton University Press,

2018.

2. Analyzing Data with Power BI and Power Pivot for Excel, Alberto Ferrari and Marco

Russo, Microsoft Press; 1st edition, 2017.

3. Microsoft Power BI Complete Reference, Devin Knight, Brian Knight, Mitchell Pearson, and Manuel Quintana, Packt Publishing; 1st edition, 2018.

Web references:

Visualization through Tableau http://www.tableausoftware.com/public

Gap Minder and Google Motion Charts (www.gapminder.org).

https://learn.microsoft.com/en-us/power-bi/

https://www.storytellingwithdata.com/